PRASAD V POTLURI SIDDHARTHA INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ACADEMIC YEAR: 2026-27

DEEP LEARNING COURSE PROJECT REPORT

MOVIE REVIEW SENTIMENT ANALYSIS USING LONG SHORT-TERM MEMORY (LSTM) NEURAL NETWORKS

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ABSTRACT

Sentiment analysis is an important application of Natural Language Processing (NLP) that identifies opinions and emotions expressed in textual data. Movie reviews provide valuable information about viewers' opinions and experiences. However, analyzing a large number of reviews manually is time-consuming. To address this problem, this project presents **Movie Review Sentiment Analysis Using Long Short-Term Memory (LSTM) Neural Networks**, a deep learning-based approach for automatically classifying movie reviews as positive or negative.

The proposed system uses the IMDb movie review dataset, which contains 50,000 labeled reviews divided into training and testing sets. The reviews are converted into numerical sequences using word indices, considering the 10,000 most frequent words. Since the reviews have different lengths, padding and truncation are applied to obtain a fixed sequence length of 200. These processed sequences are provided as input to an LSTM-based neural network.

The model consists of an embedding layer, an LSTM layer with 64 units, a dense layer with 32 neurons using the ReLU activation function, and an output layer with a sigmoid activation function. The model is compiled using the Adam optimizer and binary cross-entropy loss function. It is trained for five epochs with a batch size of 64, while 20% of the training data is used for validation. The trained model learns sequential patterns in the reviews and predicts their sentiment based on the textual content.

The performance of the model is evaluated using test accuracy, test loss, a confusion matrix, and a classification report containing precision, recall, and F1-score. Training and validation accuracy graphs are also used to observe the model's learning behavior. In addition to evaluating the model on the test dataset, the system can accept custom movie reviews and classify them as positive or negative based on the predicted sentiment score.

This project demonstrates the application of deep learning and Natural Language Processing to text classification. The developed system provides a practical approach to automated movie review analysis and can be extended in the future to support multiple sentiment categories and web-based applications.

Keywords: Sentiment Analysis, Natural Language Processing, Deep Learning, LSTM, IMDb Dataset, Text Classification.

INTRODUCTION

In recent years, online platforms have become popular sources for sharing opinions and experiences about movies. Movie reviews help people understand the quality of a movie and make informed decisions before watching it. However, analyzing a large number of reviews manually can be difficult and time-consuming. Sentiment analysis provides an automated approach to identifying opinions expressed in textual data.

Sentiment analysis is a Natural Language Processing (NLP) technique used to classify text according to the opinions or emotions it expresses. In movie review analysis, sentiment classification generally involves identifying whether a review expresses a positive or negative opinion. This process can help understand audience preferences and summarize public opinions about movies.

Deep learning techniques have become useful for text classification because they can learn patterns from large amounts of textual data. Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to process sequential information and learn relationships between words in a sequence. It uses memory mechanisms to retain relevant information while processing text, making it suitable for tasks such as sentiment analysis.

The project, **Movie Review Sentiment Analysis Using Long Short-Term Memory (LSTM) Neural Networks**, aims to develop a deep learning model that automatically classifies movie reviews as positive or negative. The IMDb movie review dataset is used to train and evaluate the model. The reviews are converted into numerical sequences and padded to a fixed length before being passed to the LSTM network.

The developed model learns patterns from the training data and predicts the sentiment of previously unseen reviews. Its performance is evaluated using accuracy, loss, a confusion matrix, precision, recall, and F1-score. The system also supports sentiment prediction for custom movie reviews provided by the user.

This project demonstrates how deep learning and Natural Language Processing can be combined to automate movie review classification. It also provides a foundation for future improvements, such as multi-class sentiment classification and the development of a user-friendly web application.

PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

Online platforms contain a large number of movie reviews expressing different opinions about movies. Manually analyzing these reviews to understand audience sentiment requires considerable time and effort. Therefore, an automated system is needed to identify whether a movie review expresses a positive or negative opinion.

The objective of this project is to develop a deep learning-based sentiment classification system using Long Short-Term Memory (LSTM) neural networks. The system uses the IMDb movie review dataset to learn patterns in textual data and classify reviews into positive and negative categories. The trained model can also predict the sentiment of new movie reviews provided by the user.

Objectives

The main objectives of this project are:

1. To understand the concepts of sentiment analysis and Natural Language Processing.
2. To use the IMDb movie review dataset for positive and negative sentiment classification.
3. To preprocess textual data by converting reviews into numerical sequences.
4. To develop an LSTM-based deep learning model for movie review classification.
5. To train the model using labeled movie reviews and evaluate its performance.
6. To measure classification performance using accuracy, precision, recall, F1-score, and a confusion matrix.
7. To predict the sentiment of custom movie reviews using the trained model.

LITERATURE REVIEW

Sentiment analysis is an important research area in Natural Language Processing (NLP) that focuses on identifying opinions expressed in textual data. Early approaches used traditional machine learning algorithms such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression. Pang, Lee, and Vaithyanathan (2002) explored machine learning techniques for sentiment classification and demonstrated their application to movie reviews. These methods commonly use text representation techniques such as Bag of Words and TF-IDF but may have limitations in capturing sequential relationships between words.

With the development of deep learning, Recurrent Neural Networks (RNNs) became useful for processing sequential data such as text. However, standard RNNs can face difficulties learning long-term dependencies. To address this limitation, Hochreiter and Schmidhuber (1997) introduced Long Short-Term Memory (LSTM) networks. LSTM uses memory cells and gating mechanisms to retain relevant information over a sequence, making it suitable for tasks such as text classification and sentiment analysis.

Maas et al. (2011), in their research paper *Learning Word Vectors for Sentiment Analysis*, explored word representations for sentiment analysis and introduced a dataset of labeled movie reviews. The IMDb movie review dataset has since been widely used for developing and evaluating sentiment classification models. It contains positive and negative reviews, making it suitable for binary sentiment classification tasks.

Deep learning-based sentiment analysis often combines an embedding layer with an LSTM network. The embedding layer converts word indices into dense vector representations, while the LSTM learns sequential patterns from the review text. A sigmoid output layer can then classify the review as positive or negative. The model's performance can be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

Based on these approaches, the present project develops an LSTM-based model for movie review sentiment classification using the IMDb dataset. The reviews are converted into numerical sequences and padded to a fixed length before being processed by the neural network. The model is trained using the Adam optimizer and binary cross-entropy loss function. Its performance is evaluated on test data, and custom reviews are used to demonstrate sentiment prediction. This project illustrates the application of deep learning to automated movie review analysis.

EXISTING SYSTEM

Existing movie review sentiment analysis systems use several approaches to identify whether a review expresses a positive or negative opinion. These include rule-based methods, traditional machine learning algorithms, and deep learning techniques.

Traditional machine learning algorithms, such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression, are commonly used for text classification. These methods often depend on text representation techniques such as Bag of Words and TF-IDF. Deep learning approaches, including Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, can learn patterns and relationships between words in review text.

These approaches are useful, but their performance depends on the quality of the dataset, text preprocessing, and the selected model. This project explores an LSTM-based approach for classifying movie reviews as positive or negative using the IMDb dataset.

LIMITATIONS OF THE EXISTING SYSTEM

The following are common limitations of existing approaches:

1. **Limited context understanding:** Traditional methods may not capture the relationships between words in a review.
2. **Dependence on text representation:** Machine learning models often require suitable text representation techniques.
3. **Difficulty with complex expressions:** Sarcasm, mixed opinions, and ambiguous statements can be difficult to classify correctly.
4. **Data dependency:** Model performance depends on the quality and diversity of the training dataset.
5. **Misclassification:** Positive reviews may be predicted as negative, and negative reviews may be predicted as positive.

These limitations motivate the use of deep learning techniques such as LSTM, which can learn sequential patterns in text. However, an LSTM model can also make incorrect predictions and must be evaluated using appropriate test data.

PROPOSED SYSTEM

The proposed system uses a Long Short-Term Memory (LSTM) neural network to classify movie reviews as positive or negative. The IMDb movie review dataset is loaded using TensorFlow/Keras, and the reviews and their corresponding sentiment labels are extracted.

The reviews are converted into numerical sequences using a vocabulary of the 10,000 most frequent words. Each sequence is padded or truncated to a fixed length of 200 words to maintain a consistent input size for the model.

An LSTM model is constructed using an embedding layer, an LSTM layer with 64 units, a dense layer with 32 neurons and ReLU activation, and a sigmoid output layer. The model is compiled using the Adam optimizer and binary cross-entropy loss function.

The model is trained for five epochs with a batch size of 64. A validation split of 20% is used to monitor the model's performance during training. After training, the model is evaluated on the test dataset using test loss and accuracy. A confusion matrix and classification report are also generated to understand its classification performance.

A prediction function is provided to classify new movie reviews as positive or negative. This allows users to enter their own reviews and observe the model's predicted sentiment.

ADVANTAGES OF THE PROPOSED SYSTEM

1. Uses an LSTM neural network to learn sequential patterns in movie reviews.
2. Converts text into numerical sequences for efficient model processing.
3. Uses an embedding layer to represent words as dense vectors.
4. Classifies movie reviews into positive and negative sentiments.
5. Uses the IMDb dataset, which contains a large collection of labeled movie reviews.
6. Evaluates model performance using test accuracy and test loss.
7. Uses a confusion matrix and classification report to analyze classification results.
8. Allows users to test the model with their own movie reviews.

SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or laptop
- Processor capable of running Python and TensorFlow
- Sufficient RAM for dataset processing and model training
- Internet connection for accessing Google Colab and loading the dataset

Software Requirements

- Python
- Google Colab
- TensorFlow and Keras
- NumPy
- Matplotlib
- Scikit-learn

Programming Language: Python

Deep Learning Framework: TensorFlow/Keras

Dataset: IMDb Movie Reviews Dataset

DATASET DESCRIPTION

This project uses the IMDb Movie Reviews Dataset, which contains movie reviews labeled as positive or negative. The dataset is loaded using TensorFlow/Keras and is divided into training and testing sets.

The dataset contains 50,000 movie reviews, with 25,000 reviews used for training and 25,000 reviews used for testing. Each review is associated with a sentiment label, making the dataset suitable for binary sentiment classification.

The dataset is loaded with a vocabulary limit of 10,000 words. The reviews are represented as numerical sequences, and each sequence is padded or truncated to a maximum length of 200 words to ensure a consistent input size.

The sentiment labels are represented in binary form:

- **Class 0:** Negative review
- **Class 1:** Positive review

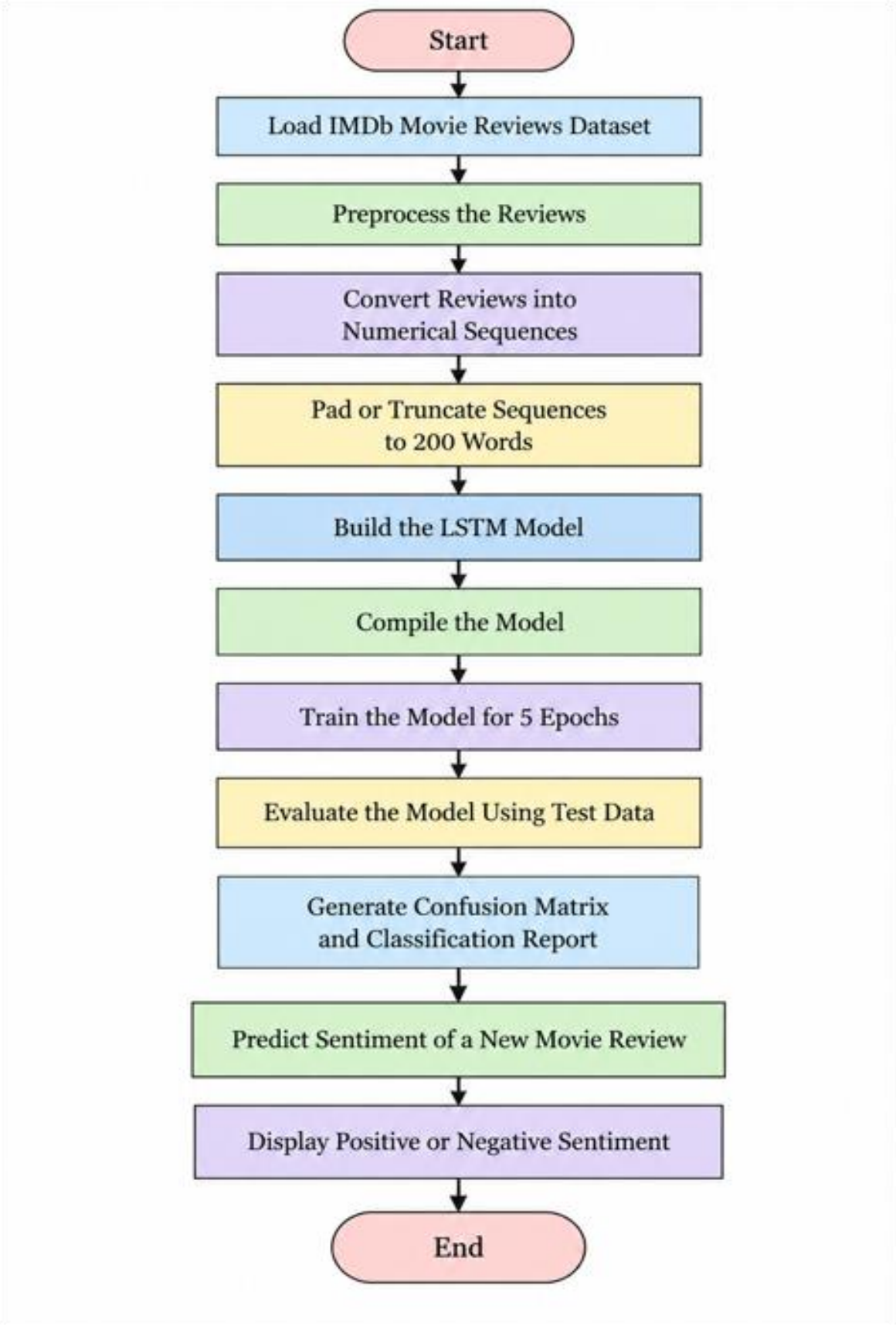
The prepared dataset is used to train the LSTM neural network and evaluate its ability to classify movie reviews into positive and negative sentiments.

METHODOLOGY

The project follows a sequence of steps, from loading the dataset to predicting the sentiment of a movie review.

1. Load the IMDb movie review dataset using TensorFlow/Keras.
2. Separate the dataset into training and testing sets.
3. Limit the vocabulary to the 10,000 most frequent words.
4. Convert the reviews into numerical sequences.
5. Pad or truncate the sequences to a maximum length of 200 words.
6. Build the LSTM neural network model.
7. Compile the model using the Adam optimizer and binary cross-entropy loss function.
8. Train the model for five epochs using a batch size of 64.
9. Use 20% of the training data for validation.
10. Evaluate the trained model using test accuracy and test loss.
11. Generate a confusion matrix and classification report to analyze the predictions.
12. Predict the sentiment of new movie reviews using the trained model.

FLOW CHART:


Start
Load IMDb Movie Reviews Dataset
Preprocess the Reviews
Convert Reviews into Numerical Sequences
Pad or Truncate Sequences to 200 Words
Build the LSTM Model
Compile the Model
Train the Model for 5 Epochs
Evaluate the Model Using Test Data
Generate Confusion Matrix and Classification Report
Predict Sentiment of a New Movie Review
Display Positive or Negative Sentiment
End

DATA PREPROCESSING

Data preprocessing converts raw movie reviews into a suitable format for the LSTM neural network.

First, the IMDb movie review dataset is loaded using TensorFlow/Keras. The dataset contains positive and negative reviews with corresponding sentiment labels. The vocabulary is limited to the 10,000 most frequent words to keep the input size manageable.

Next, each review is represented as a numerical sequence, where words are mapped to integer values based on the dataset vocabulary. Since the reviews have different lengths, the sequences are padded or truncated to a fixed length of 200 words.

Padding adds zeros to shorter sequences, while truncation removes excess words from longer sequences. This ensures that all reviews have the same input length.

The resulting numerical sequences are provided to the embedding layer of the LSTM model for sentiment classification.

DATASET SPLITTING

The IMDb movie review dataset contains 50,000 reviews, divided into 25,000 training samples and 25,000 testing samples. The dataset is split into training and testing sets to train the LSTM model and evaluate its performance on unseen data.

1. Training Dataset: The training dataset contains 25,000 movie reviews. It is used to train the LSTM model to identify patterns associated with positive and negative sentiments.

2. Testing Dataset: The testing dataset contains 25,000 movie reviews. It is used to evaluate the trained model and measure its ability to classify reviews that were not used during training.

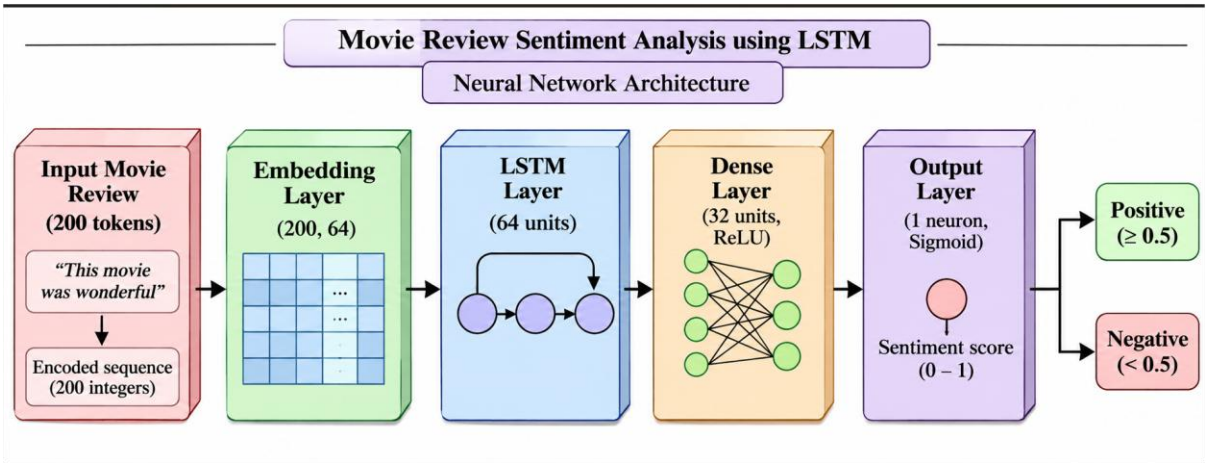
3. Validation Dataset: During training, 20% of the training dataset is used for validation. This data helps monitor the model's performance during training and compare training and validation accuracy.

The model is trained using the training data, monitored using the validation data, and evaluated using the separate testing data. This approach helps assess the model's ability to classify movie reviews correctly.

LSTM MODEL ARCHITECTURE

The proposed system uses a Long Short-Term Memory (LSTM) neural network to classify movie reviews as positive or negative. The model consists of an embedding layer, an LSTM layer, a dense layer, and an output layer.

The architecture is designed to convert numerical word sequences into meaningful representations, learn sequential patterns in the reviews, and predict the sentiment of the text.



EXPLANATION OF THE LAYERS

1. Input Layer: Accepts the preprocessed movie reviews as sequences of 200 integers. Each integer represents a word index in the dataset vocabulary.

2. Embedding Layer: Converts each word index into a learnable vector of 64 values. This helps the model learn numerical representations of words and their relationships within reviews.

3. LSTM Layer: Processes the sequence of word embeddings and learns relationships between words. It contains 64 units and captures important contextual information that helps determine the sentiment of a review.

4. Dense Layer: Contains 32 neurons with the ReLU activation function. It learns additional patterns from the features extracted by the LSTM layer.

5. Output Layer: Contains one neuron with sigmoid activation. Its output is a value between 0 and 1, representing the model's estimated positive sentiment score. A score greater than or equal to 0.5 indicates a positive review, while a score below 0.5 indicates a negative review.

MODEL TRAINING

The LSTM model is trained using the IMDb movie review training dataset, while a portion of the training data is used for validation to monitor the model's learning performance.

The model uses the Adam optimizer to update its parameters and binary cross-entropy as the loss function. Binary cross-entropy is suitable for binary classification problems involving positive and negative sentiment categories.

The training configuration specifies a maximum of 5 epochs and a batch size of 64. An epoch represents one complete pass through the training data, while the batch size determines the number of samples processed together before the model updates its parameters.

A validation split of 20% is used to monitor the model's performance during training. Training and validation accuracy can be plotted to observe the learning progress.

These settings help the model learn patterns from movie reviews and provide a basis for evaluating its classification performance.

IMPLEMENTATION DETAILS

The project is implemented in Python using TensorFlow/Keras for deep learning and Scikit-learn for model evaluation.

The implementation is divided into the following modules:

1. **Library Installation and Imports:** Loads the required Python libraries.
2. **Dataset Loading:** Retrieves the IMDb movie review dataset using TensorFlow Keras.
3. **Data Preparation:** Separates the training and testing data and prepares the review sequences.
4. **Text Preprocessing:** Limits the vocabulary to 10,000 words and pads or truncates reviews to 200 tokens.
5. **Model Construction:** Builds the embedding-based LSTM neural network.
6. **Model Compilation:** Configures the Adam optimizer, binary cross-entropy loss, and accuracy metric.
7. **Model Training:** Trains the network for 5 epochs using a batch size of 64 and a 20% validation split.
8. **Performance Evaluation:** Calculates test accuracy and test loss and generates the confusion matrix and classification report.
9. **Performance Visualization:** Plots training and validation accuracy to examine the model's learning behavior.
10. **Sentiment Prediction:** Accepts a movie review from the user and predicts whether it is positive or negative.
11. **Model Saving:** Saves the trained model in Keras format for future use.

ALGORITHM

Input: A labeled dataset containing movie reviews and their sentiment labels.

Output: A trained LSTM model that predicts whether a movie review expresses positive or negative sentiment.

Algorithm Steps:

1. Start the program.

2. Import the required libraries.
3. Load the IMDb movie review dataset.
4. Separate the training and testing datasets.
5. Select the most frequent 10,000 words.
6. Convert the reviews into numerical sequences.
7. Pad or truncate each sequence to 200 tokens.
8. Build the embedding-based LSTM neural network.
9. Compile the model using the Adam optimizer and binary cross-entropy loss.
10. Train the model for 5 epochs with a batch size of 64 and a 20% validation split.
11. Evaluate the trained model using the test dataset.
12. Calculate test accuracy and test loss.
13. Generate the confusion matrix and classification report.
14. Plot the training and validation accuracy.
15. Accept a new movie review from the user.
16. Convert the review into a numerical sequence and pad it to the required length.
17. Predict the sentiment score using the trained model.
18. Classify the review as positive or negative using a threshold of 0.5.
19. Display the predicted sentiment.

RESULTS AND DISCUSSION

The project evaluates the trained Long Short-Term Memory (LSTM) model using the IMDb movie review dataset. The model predicts whether a movie review expresses positive or negative sentiment based on its textual content.

After training, the model is evaluated using the separate test dataset. A threshold of 0.5 is used to classify the predicted sentiment scores. Reviews with a score greater than or equal to 0.5 are classified as positive, while reviews with a score below 0.5 are classified as negative.

The evaluation process calculates test accuracy and test loss to measure the model's classification performance. A classification report is generated to determine precision, recall, and F1-score for both sentiment classes. The confusion matrix provides information about correctly and incorrectly classified positive and negative reviews.

Training and validation accuracy are plotted to observe the model's learning behavior over the five training epochs. These graphs help compare performance on the training and validation data and identify possible signs of overfitting.

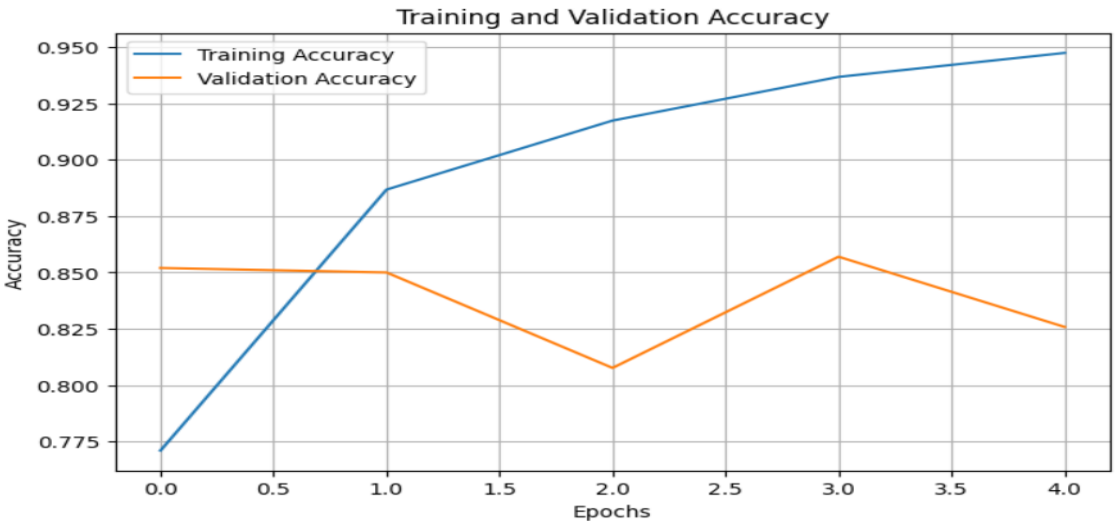
The trained model is also tested using custom movie reviews provided by the user. The system converts each review into a numerical sequence, applies padding or truncation, and predicts its sentiment using the trained LSTM model.

The final performance of the model must be reported using the actual results obtained from the notebook. The evaluation values should be copied directly from the model output rather than estimated or assumed.

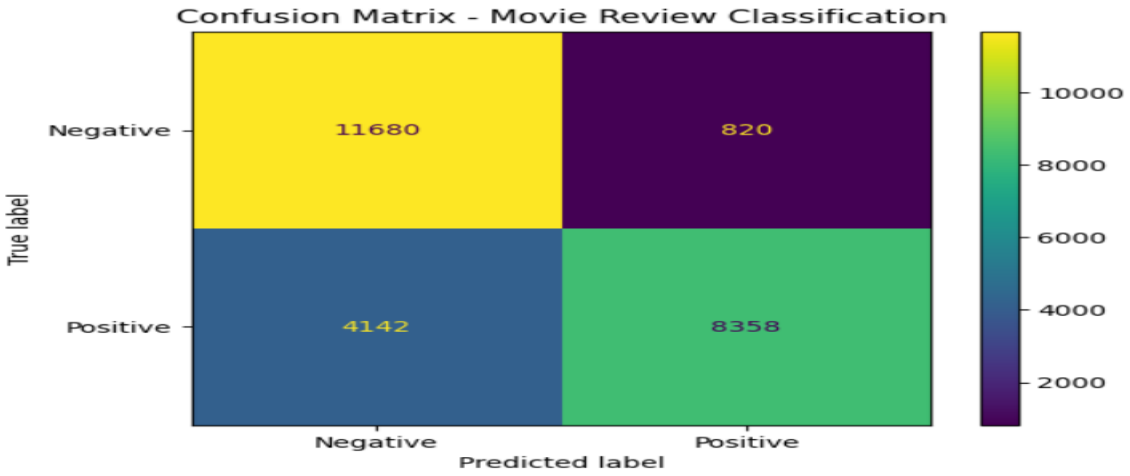
MODEL EVALUATION RESULTS

Evaluation Metric	Result
Test Accuracy	80.15%
Positive Class Precision	91.07%
Positive Class Recall	66.86%
Positive Class F1-Score	77.11%
Negative Class Precision	73.82%
Negative Class Recall	93.44%
Negative Class F1-Score	82.48%

1. **Training and validation accuracy graph.**



2. **Confusion matrix.**



3. **Screenshot of custom movie review predictions**

Enter Movie Review here

```
# Enter your movie review
review = input("Enter your movie review: ")

# Predict the sentiment
predict_sentiment(review)
```

```
Enter your movie review: predict_sentiment("Despite the excellent actors and beautiful visuals, the movie was boring and disappointing.")
Sentiment: Negative 😞
Positive sentiment score: 23.37%
```

Ente Movie Review here

```
# Enter your movie review
review = input("Enter your movie review: ")

# Predict the sentiment
predict_sentiment(review)
```

```
Enter your movie review: The movie was wonderful, entertaining, and beautifully acted.
Sentiment: Positive 😊
Positive sentiment score: 96.42%
```

The results provide an understanding of the model's ability to distinguish between positive and negative movie reviews. The actual evaluation metrics are necessary to determine how effectively the model performs this classification task.

APPLICATIONS

1. **Movie Review Classification:** Automatically classifies movie reviews as positive or negative based on their textual content.
2. **Audience Opinion Analysis:** Helps understand viewers' opinions about movies by analyzing their reviews.
3. **Movie Recommendation Platforms:** Sentiment predictions can support recommendation systems by providing additional information about viewers' opinions.
4. **Review Analysis Systems:** Can be used as a component in systems that analyze large collections of movie reviews.
5. **Entertainment Industry Research:** Helps researchers study audience feedback and sentiment patterns related to movies.
6. **Natural Language Processing Education:** Provides a practical example of applying deep learning and text classification techniques to real-world textual data.

These are potential applications of the developed model. The current implementation focuses on classifying movie reviews and predicting their sentiment. It does not independently provide a complete movie recommendation platform or commercial review analysis system.

ADVANTAGES

- Automatically classifies movie reviews into positive and negative categories.
- Uses an LSTM network to learn sequential patterns in textual data.
- Uses an embedding layer to convert numerical word indices into dense vector representations.
- Applies padding and truncation to process reviews of different lengths.
- Uses the Adam optimizer and binary cross-entropy loss for model training.
- Evaluates performance using accuracy, precision, recall, F1-score, and a confusion matrix.
- Supports sentiment predictions for user-entered movie reviews.
- Allows the trained model to be saved and reused later.

LIMITATIONS

- The model classifies reviews only as positive or negative.
- The results depend on the quality and characteristics of the training dataset.
- The vocabulary is limited to the 10,000 most frequent words.

- Reviews longer than 200 tokens are truncated, so information beyond that limit is not considered.
- The model may have difficulty understanding sarcasm, irony, and complex expressions.
- The model may perform differently on reviews from other datasets or domains.
- Training a deep learning model requires computational resources and time.
- The predicted sentiment score is a model estimate and may not represent a calibrated probability of the review's true sentiment.

FUTURE ENHANCEMENTS

The project can be extended in several ways:

1. **Web Application Development:** Develop a user-friendly web application where users can enter movie reviews and view their predicted sentiment.
2. **Multi-Class Sentiment Classification:** Extend the model to classify reviews into multiple categories, such as very positive, positive, neutral, negative, and very negative, using an appropriate labeled dataset.
3. **Dataset Expansion:** Train and evaluate the model using additional movie review datasets to improve its evaluation across different types of reviews.
4. **Model Comparison:** Compare the LSTM model with CNN, GRU, and transformer-based models.
5. **Hyperparameter Optimization:** Experiment with different embedding dimensions, LSTM units, batch sizes, and training configurations.
6. **Improved Text Processing:** Explore preprocessing techniques to better handle punctuation, unfamiliar words, and complex expressions.
7. **Model Optimization:** Investigate methods to reduce model size and prediction time.
8. **Explainable Predictions:** Introduce techniques to help users understand which aspects of a review may have influenced its predicted sentiment.
9. **Recommendation System Integration:** Combine sentiment predictions with user preferences and movie information to support a movie recommendation system.
10. **Generalization Evaluation:** Evaluate the model on additional datasets to study its performance on reviews that differ from the original training data.

CONCLUSION

This project presents a movie review sentiment classification system using a Long Short-Term Memory (LSTM) neural network. The system uses the IMDb movie review dataset, applies text preprocessing

techniques, converts reviews into numerical sequences, and trains a deep learning model to classify reviews as positive or negative.

The model architecture consists of an embedding layer, an LSTM layer, a dense layer with ReLU activation, and an output layer with sigmoid activation. The model is trained using the Adam optimizer and binary cross-entropy loss function. Padding and truncation are used to ensure that the input reviews have a fixed sequence length of 200 tokens.

The implementation evaluates the model using test accuracy, test loss, precision, recall, F1-score, and a confusion matrix. Training and validation accuracy graphs are also used to observe the model's learning behavior. In addition, the system supports sentiment prediction for custom movie reviews.

The project demonstrates the application of deep learning and Natural Language Processing to automated text classification. Although the model provides a practical approach to identifying positive and negative movie reviews, further improvements and testing on diverse datasets may help enhance its performance and generalization.

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